Clinical science

Drusen volume and reticular pseudodrusen volume from optical coherence tomography with deep learning as risk factors for progression to late age-related macular degeneration in eyes with reticular pseudodrusen and contralateral macular neovascularisation

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ABSTRACT

Aim To implement a deep learning-based segmentation algorithm to quantify reticular pseudodrusen (RPD) and drusen volumes on optical coherence tomography (OCT) and investigate their association with progression to late age-related macular degeneration (AMD).

Methods A retrospective analysis included study eyes with RPD and contralateral neovascular AMD using 6×6 mm macular OCT (Solix; Visionix/Optovue, Inc). Automated segmentation quantified RPD and drusen volumes, including large drusen and drusenoid pigment epithelial detachment (PED), and late AMD development was evaluated over 2 years. Associations between baseline volumetric biomarkers and progression were evaluated using Cox proportional hazards models.

Results Fifty-one eyes (mean age 74.9±7.65 years) were included. The median (IQR) baseline RPD volume was 0.018 mm³ (0.004–0.52) and total drusen volume was 0.009 mm³ (0.001–0.064). Over 24.2±1.20 months, late AMD developed in 20 eyes (39.2%). In multivariable Cox regression models adjusted for age, each 0.01 mm³ increase in baseline RPD volume (HR: 1.082; p=0.002) and total drusen volume (HR: 1.080; p<0.001) were associated with progression to late AMD. In an exploratory drusen subtype analysis, progression to late AMD was independently associated with baseline RPD volume, large drusen volume and drusenoid PED volume.

Conclusion The deep learning-based volumetric segmentation tool allows OCT-derived automated volume quantification of different types of drusen based on OCT. In eyes with RPD and contralateral neovascular AMD, greater RPD volume and large drusen and/or drusenoid PED volume carried greater risk of developing late AMD in 2 years.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Reticular pseudodrusen (RPD) is a significant risk factor for progression to late age-related macular degeneration (AMD), particularly when coexisting with conventional drusen. However, prior studies have predominantly relied on binary grading of RPD presence or absence, rather than quantitative assessment that accounts for concurrent drusen or drusenoid pathology.

WHAT THIS STUDY ADDS

⇒ This study demonstrates that a deep learning-based automated volumetric analysis of RPD and drusen subtypes on optical coherence tomography (OCT) enables accurate differentiation and quantification of clinically relevant lesions, and that greater baseline RPD volume and large drusen and/or drusenoid pigment epithelial detachment volume predict progression to late AMD in eyes with RPD and contralateral neovascular AMD within 2 years.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Currently, quantitative drusen volume analysis is not used in routine clinical practice. Integrating automated volumetric quantification of RPD and drusen subtypes into AMD grading systems may increase accessibility to OCT-based biomarkers, improve risk stratification and strengthen disease monitoring in clinical practice.

INTRODUCTION

Age-related macular degeneration (AMD) is a leading cause of irreversible vision impairment among the elderly, with late-stage complications, including geographic atrophy (GA) and macular neovascularisation (MNV), contributing to progressive and severe vision loss.[1 2] Drusen, the hallmark early feature of AMD, have long been recognised as a major risk factor for disease progression.[3] However, emerging evidence suggests that reticular pseudodrusen (RPD), or subretinal drusenoid deposits,[4] represent a distinct and clinically significant phenotype of AMD.[5] A recent post

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hoc analysis of the Age-Related Eye Disease Study (AREDS) and AREDS2 cohorts demonstrated that RPD is an important risk factor for progression to late AMD, and emphasised the need to integrate RPD into AMD classification systems and risk prediction models.[6] However, the predictive features for disease progression were limited by the imaging modality used, as analyses were based on colour fundus photography (CFP) and relied on a binary classification of presence or absence, rather than quantitative metrics of RPD status. Further, inter-reader variability in assessing RPD on optical coherence tomography (OCT) volume scans and B-scans was substantial but not perfect, highlighting challenges in manual grading and its potential impact on the consistency of clinical association studies.[7]

Our group has demonstrated that an automated approach achieves an accurate segmentation and quantification of three-dimensional RPD and drusen volumes from OCT scans.[8 9] Thus far, only a limited number of studies have explored deep learning techniques for the automatic detection of RPD on OCT.[10 11] A recent study by Kumar *et al* developed a deep learning model for RPD segmentation in AMD and demonstrated robust performance across multiple large, externally validated datasets.[12] However, these approaches focused on segmentation performance and did not evaluate quantitative RPD outputs as biomarkers for clinical risk assessment. Furthermore, our algorithm successfully segments and quantifies different volumetric drusen subtypes, including RPD and drusenoid pigment epithelial detachment (PED), offering a more precise risk stratification for AMD progression. This approach may enhance our understanding of the relative contributions of RPD and drusen subtypes in disease progression and their inter-relationships.

In this study, we selected eyes for AMD progression based on risk factors identified in the Beckman Initiative for Macular Research Classification.[2] Using deep learning-based segmentation and quantification of RPD and drusen volume at baseline, we longitudinally followed fellow eyes with RPD in patients with exudative neovascular AMD to evaluate the rate progression to late AMD disease over a 2-year period.

METHODS

Study participants

The data reported in this retrospective cohort study comprises cross-sectional analyses from a prospective observational study (NIH R01-EY024544) at the Casey Eye Institute, Oregon Health & Science University, from February 2019 through May 2024. The study adhered to the tenets of the Declaration of Helsinki and complied with the Health Insurance Portability and Accountability Act of 1996.

The study enrolment criteria included subjects aged 50 years or older, study eyes required evidence of RPD with or without drusen on clinical examination in patients with unilateral history of neovascular AMD. All patients had a diagnosis of MNV in the fellow eye and were treated with various anti-vascular endothelial growth factor (VEGF) agents. Exclusion criteria for the study eyes included the presence of OCT-defined atrophy (from nascent GA[13] or worse, including any complete retinal pigment epithelium (RPE) and outer retinal atrophy (cRORA)[14]) and late AMD at baseline. Eyes with serous or fibrovascular PED caused by sub-RPE fluid, blood or fibrovascular material, as well as those with concurrent retinal pathologies such as diabetic retinopathy, macular nevus or a history of vitrectomy, were excluded. All cases were retrospectively reviewed to confirm eligibility and diagnosis. Verification of inclusion and exclusion criteria was performed through direct review of imaging and review of relevant clinical records to assess ocular and systemic history. Additionally, eyes with images of poor image quality due to motion artefacts or inadequate signal strength less than 50 were also excluded.

Image acquisition and analysis

All participants underwent a medical history inquiry, comprehensive ophthalmic examinations, including standard Early Treatment Diabetic Retinopathy Study (ETDRS) visual acuity testing, intraocular pressure measurement, slit-lamp biomicroscopy and dilated fundus examination. Imaging was performed using 6×6 mm OCT volumetric scans with a 400×400 or 512×512 sampling density from study eye were obtained using a commercial 120 kHz spectral domain-OCT system (Solix; Visionix/Optovue, Inc, California, USA). The clinical presence of RPD was defined as hyporeflective lesions with or without target configuration against a background of mild hyper-reflectance on near-infrared on confocal scanning laser ophthalmoscopy with corresponding hyper-reflective signal above the retinal pigment epithelium on structural OCT.[4 15] In accordance with previously published criteria, only stage 2 or 3 RPD lesions were considered, defined as mounds of accumulated material sufficient to alter the contour of the ellipsoid zone boundary or lesions in which the material adopted a conical appearance and breached the ellipsoid zone.[4] RPD presence was defined by the detection of five or more RPD lesions identified on more than one B-scan.[5] No clustering requirement was applied.

To perform image processing for baseline RPD and drusen quantification, a deep-learning algorithm was employed for automated segmentation and volumetric quantification of RPD and drusen on 6×6 mm-volumetric OCT scans.[9] The deep learning model was developed using OCT images from a separate group of participants and did not include images from eyes analysed in the present longitudinal study, although both were acquired at the same institution using the same OCT device. In our segmentation algorithm, quantitative metrics of RPD were identified and measured separately, as they are not classified as drusen. Total drusen volume (excluding RPD) was defined as the sum of all drusen volumes regardless of size. For further stratification, drusen were categorised based on the maximum basal diameter, measured two-dimensionally from the *en face* projection of the OCT volume. Based on this measurement, drusen were classified as small and/or medium drusen (maximum basal diameter ≤125 µm), large drusen (maximum basal diameter >125 µm and ≤350 µm) and drusenoid PED (maximum basal diameter >350 µm), defined as a large druse or a confluence of drusen, as previously defined by AREDS.[2 16] The combined large drusen/drusenoid PED volume was calculated as the sum of large drusen volume and drusenoid PED volume. The accuracy and validity of this deep-learning-based approach for RPD and drusen volume measurement have been described previously.[9] The input consists of volumetric OCT images, which are processed by a convolutional neural network to segment the RPE–drusen complex and drusenoid volume. The output includes labelled segmentations of different drusen types including pseudodrusen, drusen and drusenoid PED, based on their anatomical location and size characteristics. During preprocessing, a denoising step using a three-voxel filter was applied to remove very small signals in both drusen and RPD segmentations.

End points

All participants were followed for a period of 2 years, and the time to the development of late AMD was determined. Study

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visits occurred at approximately 3-month intervals (±1 month), corresponding to serial follow-up visits over the 2-year observation period. Late AMD was classified as either cRORA or MNV depending on whichever late stage developed first and was not further evaluated for a second late-stage event. cRORA was considered present when the diagnostic criteria were met on any B-scan within the OCT volume. MNV was defined and diagnosed based on clinical examination, with confirmation by OCT, OCT angiography and fluorescein angiography, if necessary. All cases were retrospectively reviewed to confirm diagnosis. Grading was independently performed by two masked graders (A-LW and YG), with confirmation by an expert grader (STB).

Statistical analysis

Descriptive statistics were computed for baseline characteristics, with continuous variables summarised as means and SD and categorical variables presented as proportions. Time-to-event analysis was performed using Kaplan-Meier survival curves to estimate the cumulative incidence of late AMD. Cox proportional hazards models were used to evaluate the association between baseline volumetric biomarkers and progression to late-stage AMD over the 2-year follow-up period in univariable and multivariable models. Two multivariable models were constructed to account for different combinations of drusen subtypes. Model 1 included baseline age, RPD volume and total drusen volume, which are recognised factors associated with AMD progression.[2 6 16 17] Model 2 included baseline age, RPD volume, large drusen volume (excluding drusenoid PED volume) and drusenoid PED volume to explore morphologic subtype volume with AMD progression. All statistical tests were two-sided and p values <0.05 were considered statistically significant. Data were analysed using SPSS Statistics (V.26.0; IBM Corp, Armonk, New York, USA).

RESULT

Study cohort and baseline characteristics

A total of 51 eyes from 51 participants were included in the analysis. The mean age of participants was 74.9±7.65 years (range 56–91 years), 32 (62.7%) were women and 19 (37.3%) were men. Thirty-seven (72.5%) patients were noted to have hypertension, 7 (13.7%) patients had diabetes without retinopathy and 23 (45.1%) had a history of cardiovascular diseases. At baseline, all study eyes exhibited evidence of RPD, with varying degrees of drusen burden. The median (IQR) baseline RPD volume was 0.018 mm³ (0.004–0.52), while the median (IQR) total drusen volume (excluding RPD) was 0.009 mm³ (0.001–0.064). The correlation between RPD volume and total drusen volume is shown in figure 1. The Pearson correlation coefficient was r=0.350. A total of 11 (21.6%) eyes had only small/medium drusen, 39 (76.5%) eyes had large drusen (6 without drusenoid PED) and 34 (66.7%) eyes exhibited drusenoid PED (1 without large drusen). The volumes of baseline large drusen and drusenoid PED are detailed in table 1.

Progression to late AMD

Over the mean follow-up of 24.2±1.20 months period, 20 (39.2%) out of the full cohort (51 eyes) progressed to late AMD, with cRORA developing in 8 (15.7%) eyes and MNV in 12 (23.5%) eyes (figure 2). Kaplan-Meier cumulative incidence plots stratified by the presence or absence of coexisting large drusen and/or drusenoid PED demonstrated differences in time to progression (figure 3). Quantitative associations were subsequently evaluated using volumetric univariable and multivariable Cox proportional hazards models. In univariable analyses, higher baseline RPD, total drusen, large drusen and drusenoid PED volumes were each significantly associated with progression to late AMD over the 2-year follow-up period. In multivariable Cox proportional hazards models adjusted for age, both baseline RPD volume and total drusen volume remained independently associated with progression to late AMD. Specifically, each 0.01 mm³ increase in RPD volume was associated with a HR of 1.082 (95% CI 1.029 to 1.138; p=0.002), while each 0.01 mm³ increase in total drusen volume was associated with an HR of 1.080 (95% CI 1.032 to 1.131; p<0.001). In exploratory morphological drusen subtype

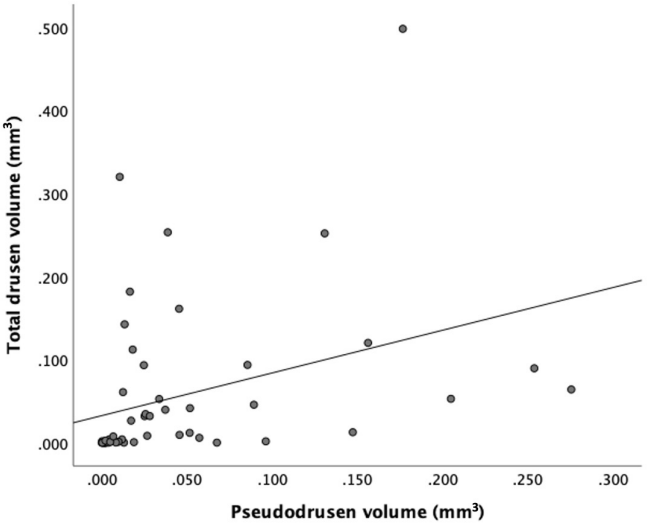


Figure 1 Scatter plot illustrating the relationship between reticular pseudodrusen volume and total drusen volume (Pearson correlation coefficient r=0.350). Each dot represents one eye.

Table 1 Baseline demographic and clinical characteristics of eyes included in the study

	All eyes (n=51)
Age (range), years	74.9±7.65 (56–91)
Sex (%)	
Male	19 (37.3)
Female	32 (62.7)
Laterality (%)	
Right	27 (52.9)
Left	24 (47.1)
Medical history (%)	
Hypertension	37 (72.5)
Diabetes	7 (13.7)
Cardiovascular disease	23 (45.1)
Baseline OCT characteristics	
Pseudodrusen volume (mm³), median (IQR)	0.018 (0.004–0.052)
Total drusen volume (mm³), median (IQR)	0.009 (0.001–0.064)
Large drusen volume (mm³), median (IQR)	0.003 (0.000–0.013)
Drusenoid PED volume (mm³), median (IQR)	0.003 (0.000–0.044)
Baseline BCVA, ETDRS letter score	79.1±7.05
Final BCVA, ETDRS letter score	79.5±6.32
Follow-up months	24.2±1.20

AMD, age-related macular degeneration; BCVA, best-corrected visual acuity; ETDRS, Early Treatment Diabetic Retinopathy Study; OCT, optical coherence tomography; PED, pigment epithelial detachment.

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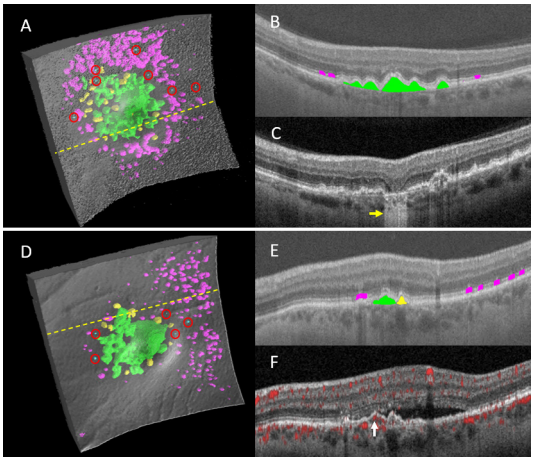


Figure 2 Baseline detection of RPD and drusen that progressed to (A–C) cRORA and to (D–F) MNV over a 2-year follow-up. (A) Three-dimensional volume-rendered OCT at baseline showing RPD (magenta), small and medium drusen (blue; marked with red circles), large drusen (yellow) and drusenoid PED (green). (B) Corresponding cross-sectional OCT at the location indicated by the yellow dotted line in (A). (C) At the 2 year follow-up, cRORA developed at the corresponding location, accompanied by choroidal hypertransmission (yellow arrow). (D) Baseline volume-rendered OCT with RPD, small and medium drusen, large drusen and drusenoid PED colour-coded as magenta, blue, yellow and green, respectively. (E) Cross-sectional OCT at the location indicated by the yellow dotted line in (D). (F) At the 2-year follow-up, MNV developed, as shown by the OCT B-scan with flow overlay, demonstrating abnormal flow signals in the sub-RPE space (white arrow) and associated subretinal fluid. cRORA, complete RPE and outer retinal atrophy; MNV, macular neovascularisation; OCT, optical coherence tomography; PED, pigment epithelial detachment; RPD, reticular pseudodrusen; RPE, retinal pigment epithelium.

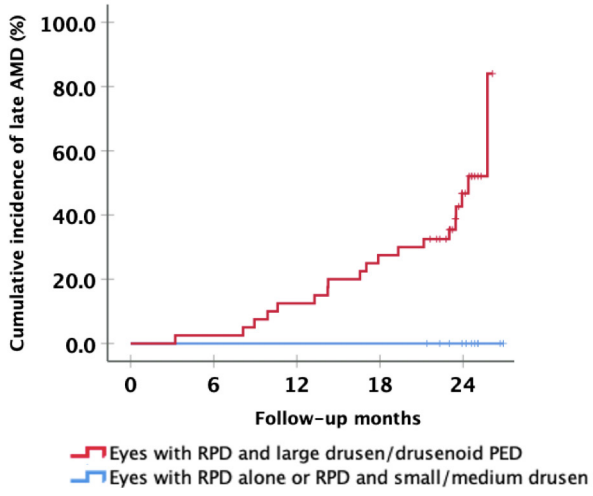


Figure 3 Kaplan-Meier cumulative incidence curves showing progression to late AMD. Cumulative incidence was derived as 1 minus the Kaplan-Meier survival probability. Eyes were stratified by the presence of RPD with large drusen or drusenoid PED (red line) versus RPD alone or RPD with only small to medium drusen (blue line). Eyes with RPD alone or with coexisting with drusen smaller than large, did not develop late AMD. AMD, age-related macular degeneration; PED, pigment epithelial detachment; RPD, reticular pseudodrusen.

analyses, volumetric biomarkers of RPD, large drusen and drusenoid PED also remained independently associated with progression to late AMD. Specifically, each 0.01 mm^3 increase in RPD volume was associated with an HR of 1.057 (95% CI, 1.002 to 1.114; p=0.043). For large drusen and drusenoid PED, each 0.01 mm^3 increase in volume was associated with HRs of 1.925 (95% CI 1.085 to 3.415; p=0.025) and 1.070 (95% CI 1.022 to 1.121; p=0.004), respectively. Baseline age was also independently associated with progression to late-stage AMD, with an HR of 1.076 per year increase (95% CI 1.001 to 1.156; p=0.048). Univariable and multivariable Cox proportional hazards results for all baseline variables are summarised in table 2. An additional analysis was conducted for each of the two models using cube root-transformed volumetric analysis (online supplemental tables S1 and S2). Results for RPD and total drusen cube root-transformed volumetric analysis (model 1) were consistent with the volumetric analysis. Cube root-transformed volumetric analysis for RPD, large drusen and drusenoid PED (model 2) had a similar trend as the corresponding volumetric analysis but results only reached borderline statistical significance.

DISCUSSION

RPD have historically been difficult to detect in routine clinical practice due to their subtle appearance on clinical examination when assessed solely with CFP, as they may resemble and be mistaken for other AMD phenotypes, such as conventional drusen.[18 19] Histologically, RPD differ from drusen as they are located above the RPE and have distinct compositional characteristics, despite sharing some similar components with drusen, suggesting different pathogenic pathways.[20 21] Multiple population-based and research cohort studies have reported varying prevalence rates of RPD in early and intermediate AMD, largely influenced by differences in imaging modalities and definitions used.[22–24] Given that histopathological evidence supports the characteristic RPD findings observed on OCT imaging,[4] our group has developed deep learning algorithms for RPD segmentation, enabling precise tracking of individual RPD lesions rather than relying solely on clustering patterns.[9] This approach improves efficiency while maintaining good sensitivity and specificity, and has the potential to be integrated into routine clinical workflows as a screening tool for AMD risk assessment.

Our findings are consistent with previous studies on the prognostic significance of RPD in the fellow eyes of individuals with unilateral neovascular AMD, demonstrating that RPD is a key risk factor for AMD progression, particularly in the conversion to late-stage disease.[25 26] In the Comparison of AMD Treatment Trials (CATT) study, Zhou *et al* reported that late AMD developed in 81 (46%) eyes where RPD was detected via fundus photography and fluorescein angiography over a 2-year follow-up period. Additionally, the presence of RPD was associated with a 2.1-fold increased risk of late AMD progression, after adjusting for large drusen and pigmentary abnormalities.[26] More recently, the HARBOR trial found that fellow eyes with OCT-defined RPD had a 2.0-fold increased risk of developing late AMD over 24 months compared with eyes without RPD, with 33.13% of eyes progressing to cRORA or MNV, after adjusting for additional OCT-based AMD risk factors.[25] In these studies, RPD presence was assessed using binary method, without a clear understanding of how RPD quantity influences disease risk. By quantifying RPD volumetric burden, including various subtypes,[15] our study enables a more refined risk stratification beyond the limitations of conventional binary classifications.

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Table 2 Association of baseline characteristics with progression to late-stage AMD

	Univariable			Multivariable model 1			Multivariable model 2		
Baseline variables	HR	95% CI	P value	HR	95% CI	P value	HR	95% CI	P value
Age (per year)	1.018	0.961 to 1.079	0.548	1.054	0.982 to 1.132	0.146	1.076	1.001 to 1.156	0.048
Sex (female vs male)	0.765	0.317 to 1.848	0.552						
Hypertension (present vs absent)	0.519	0.211 to 1.277	0.153						
Diabetes (present vs absent)	0.775	0.178 to 3.374	0.734						
Cardiovascular disease (present vs absent)	1.045	0.430 to 2.542	0.922						
Pseudodrusen volume (per 0.01 mm^3)	1.099	1.050 to 1.151	<0.001	1.082	1.029 to 1.138	0.002	1.057	1.002 to 1.114	0.043
Total drusen volume (per 0.01 mm^3)	1.081	1.042 to 1.121	<0.001	1.080	1.032 to 1.131	<0.001			
Large drusen volume (per 0.01 mm^3)	1.961	1.370 to 2.807	<0.001				1.925	1.085 to 3.415	0.025
Drusenoid PED volume (per 0.01 mm^3)	1.079	1.039 to 1.121	<0.001				1.070	1.022 to 1.121	0.004

AMD, age-related macular degeneration; PED, pigment epithelial detachment.

While drusen size and extent have long been recognised as critical risk factors for AMD progression, our findings underscore the importance of volumetric assessment in providing a more comprehensive evaluation of disease burden. Previously, an OCT-derived central drusen volume of ≥0.03 mm^3 has been identified as a risk factor for progression to late AMD in eyes with unilateral neovascular AMD.[25 27] Building on the size-based risk stratification framework originally defined using CFP, our study further categorises drusen into large drusen and drusenoid PED subgroups based on basal diameter derived from *en face* projection of OCT volume, adhering to the established standardised definitions.[2 16 19] Although drusen size metrics derived from CFP and OCT are not directly interchangeable, and measurement discrepancies may exist due to uncorrected differences between modalities, prior studies have demonstrated that OCT can reliably classify drusen according to size categories established by CFP.[28] Our findings support the established role of total drusen volume as an important measure of overall drusen burden. In addition, volumetric assessment of large drusen and drusenoid PED may provide insight into the composition of drusen pathology. This exploratory approach in this study may help refine OCT-based AMD grading frameworks. However, larger studies are needed to determine whether individual drusen morphologies provide additional prognostic information beyond total drusen volume.

The presence of RPD is well established as an additional macular risk feature for progression to late AMD.[29] However, it is not considered a totally independent risk factor, in part because their additional risk is not distributed evenly across the spectrum of nonadvanced AMD severity.[6] Within existing framework, binary assessment of RPD presence has been shown to be sufficient for broad risk categorisation while preserving clinical simplicity, but such approach does not address whether the magnitude of RPD burden provides prognostic information independent of coexisting drusen pathology.[6 29 30] In the present study, we applied deep learning-based automated segmentation to quantify both RPD and drusen as continuous volumetric measures and evaluated their effects concurrently within a multivariable framework. After adjustment for age and total drusen volume, RPD volume remained significantly associated with progression, suggesting that quantitative assessment of RPD burden may capture clinically meaningful risk gradients not fully reflected by binary RPD classification. We intentionally restricted the cohort to eyes with established RPD to evaluate the interaction between RPD and coexisting drusen burden. Accordingly, this study represents an initial step towards burden-based risk stratification of RPD, and future studies incorporating eyes with and without RPD will be required to more comprehensively assess the independent contribution of RPD across the broader spectrum of non-advanced AMD.

Although multimodal imaging remains the gold standard for comprehensive AMD assessment, our findings highlight the practical utility of OCT alone in simultaneously detecting and quantifying RPD and drusen. In particular, the coregistered structural OCT with higher sampling density and angiographic images derived from a single OCT angiography scan enable efficient evaluation of both structural and vascular biomarkers,[31] including RPD, drusen, fluid and MNV.[32] Importantly, this integrated imaging approach with automated quantification enhances accessibility and clinical usability of OCT-based biomarkers by embedding highly repeatable analysis into routine workflows, supporting more accurate AMD classification and risk stratification.[8 9 33 34] In addition, automated longitudinal assessments may facilitate earlier detection of disease progression and allow more personalised follow-up strategies.

There are several limitations to this study including a small sample size and retrospective study design. Second, the study cohort consisted of eyes with contralateral neovascular AMD, and therefore the findings may not be generalisable to patients with non-advanced AMD in both eyes. Third, our volumetric analysis used the entire OCT macular cube, rather than assessing lesions within the AMD classification-defined region of two-disc diameters from the fovea.[2] Fourth, RPD and drusen volumes can fluctuate over time, as lesions appear and regress, posing challenges when using a single time-point volumetric measurement for patient risk stratification.[35] However, the study cohort was observed over a fixed follow-up period, which helps mitigate potential variability and its impact on the analysis. Fifth, our algorithm did not consider internal reflectivity or drusen homogeneity, which may represent different biological conditions within drusen. Sixth, our study did not account for the possible effects of AREDS supplementation and smoking which are two key modifiable risk factors associated with AMD progression.[36–38] Lastly, this study used cRORA-based OCT criteria as a surrogate for advanced atrophic AMD, instead of GA based on CFP. Nascent GA is an alternative OCT based surrogate for GA, though our study did not evaluate this as an endpoint. The rates of progression to GA are different for cRORA and nascent GA and each have different responses to microperimetry.[39–41] Accordingly, the use of cRORA alone as a surrogate for

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end-stage atrophic AMD may have inherent limitations. While OCT-based definitions likely provide a more comprehensive representation of AMD-related structural changes, these definitional differences should be considered.

In conclusion, this study highlights the prognostic value of deep learning-based volumetric segmentation in assessing RPD and drusen for AMD progression. By quantifying lesion burden from OCT alone, we identified baseline RPD and drusen volumes as significant predictors of progression to late AMD within 2 years in eyes with RPD and contralateral neovascular AMD. These findings support integrating volumetric analysis into an OCT-based AMD grading system, enhancing risk stratification and disease monitoring.

Contributors I confirm all authors meet the four ICMJE criteria for authorship and are author contributors. STB is the guarantor.

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Competing interests YG: Optovue/Visionix (P), Genentech/Roche (P, R), Ifocus Imaging (P); TTH: Ifocus Imaging (I); DH: Optovue/Visionix (F, P, R), Boeringer Ingelheim (C), Genentech/Roche (P, R), Cylite (F), Intalight (F), Kugler (R), Elsevier (R), GoCheck Kids (I, P), Stroma (I), Unfold Therapeutics (I, P); YJ: Optovue/Visionix (P, R), Genentech/Roche (P, R, F), Ifocus Imaging (I, P), Optos (P), Boeringer Ingelheim (C), Kugler (R); STB: Optovue/Visionix (F). These potential conflicts of interest have been reviewed and managed by OHSU. Other authors declare no conflicts of interest related to this article.

Patient consent for publication Not applicable.

Ethics approval Ethical approval was obtained from the Institutional Review Board of Oregon Health & Science University (IRB#25487). Written informed consent was obtained from all participants prior to enrolment. Participants gave informed consent to participate in the study before taking part.

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